

Derbyshire Shared Care Pathology Guidelines

Abnormal Liver Function Tests (LFTs) in Adults



Dependent upon clinical details (when given) and previous LFT results the laboratory usually telephones the following abnormal results Monday to Friday when the GP practices are open:

- **Albumin < 15g/L**
- **Bilirubin > 200 µmol/L**
- **ALT > 300 IU/L**

In patients where the LFTs are persistently grossly abnormal, results are not generally telephoned through to the practice.

Dependent upon clinical details (when given) and previous LFT results the laboratory may telephone **ALT results ≥ 600 IU/L** to Derbyshire Health United when the GP practice is closed.

Further investigation of abnormal liver function tests

It is important to note that there are many causes of transient elevation of liver enzymes including inter-current illness. The following decision trees on pages 2 & 3 of this document outline the appropriate further investigation and referral of patients with abnormal LFTs.

Important questions to consider are:

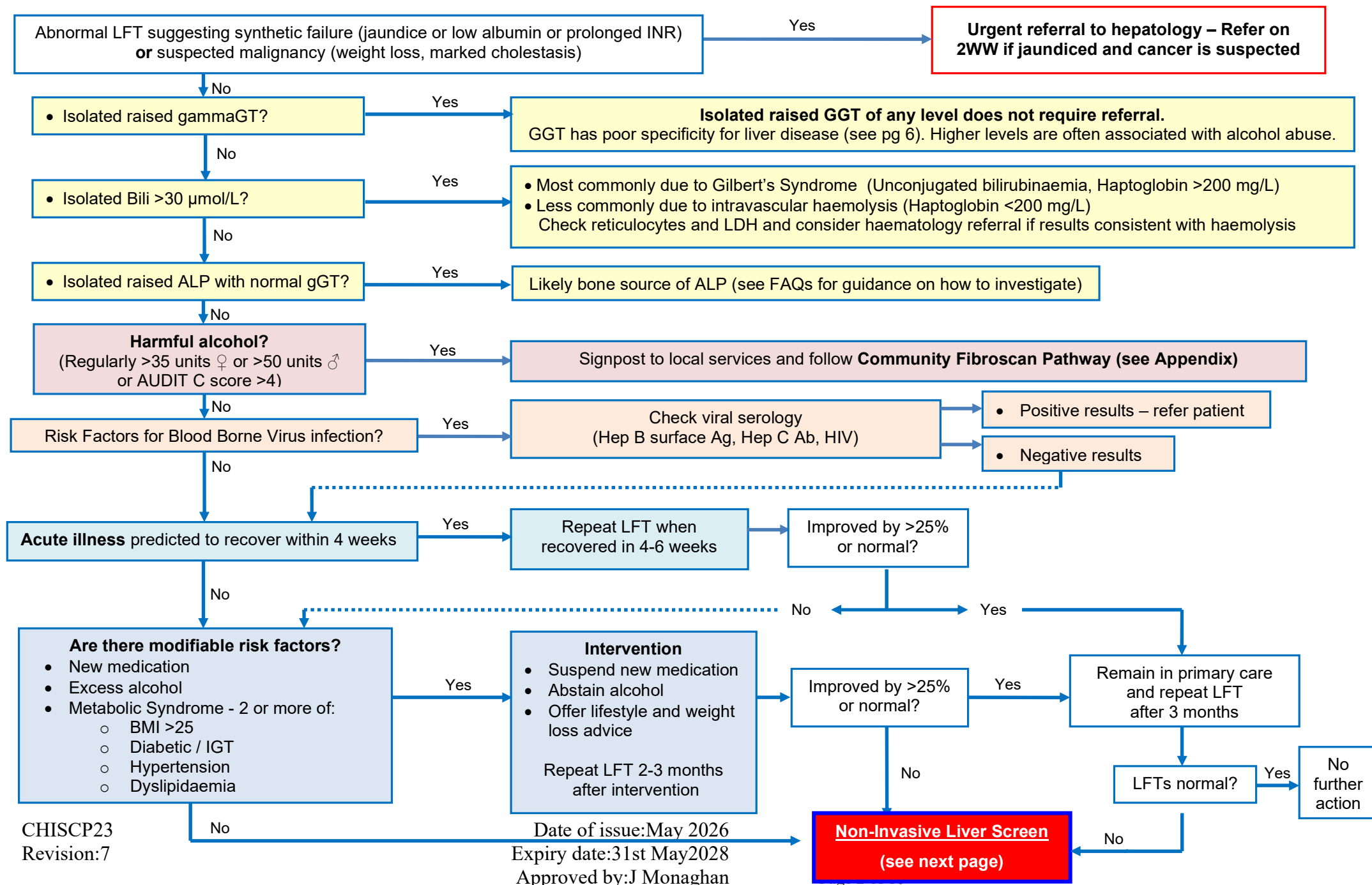
- Are the deranged LFTs chronic or an acute change?
- Are there signs of liver failure which should prompt urgent assessment?
- Are the abnormal LFTs associated with severe pain which should prompt urgent assessment, particularly if any signs of sepsis?

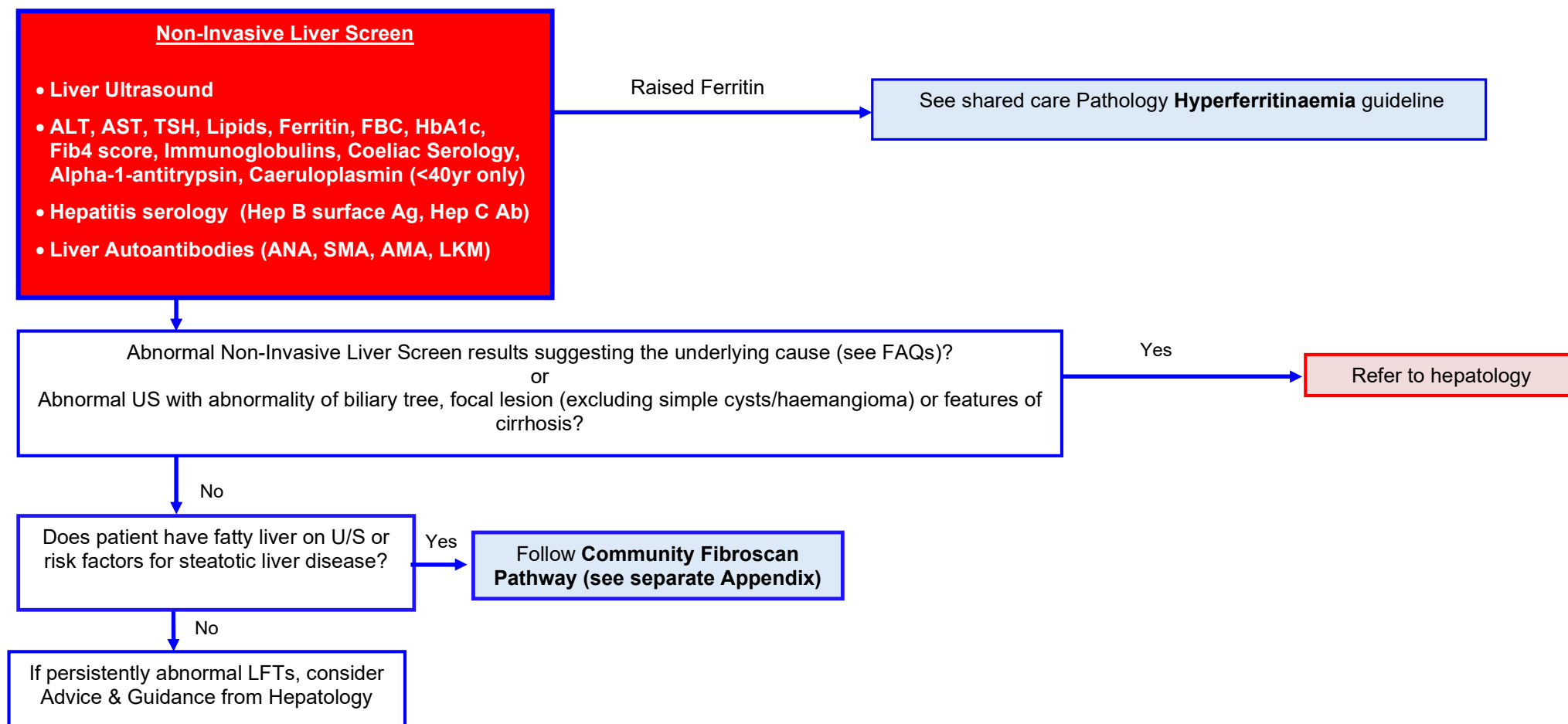
A full liver screen IS NOT REQUIRED IMMEDIATELY in:

- SYMPTOMATIC patients there is an alternative explanation – e.g. intercurrent illness or medication
- Patients with modifiable risk factors (metabolic syndrome, alcohol)

Prior to referral it is important to have first carried out a Non-Invasive Liver Screen (NILS) as outlined in the decision tree, to aid efficient patient management.

Background information on individual parameters on the LFT profile follows on pages 6-8.





Frequently Asked Questions

- **How do I interpret an abnormal alpha-1-antitrypsin level?**

When the alpha₁-antitrypsin level is < 1.1g/L, phenotype analysis will be performed (if not previously done). Please refer patients with PiZZ phenotype or PiSZ if the patient also has other risk factors (e.g. alcohol). For more information, please visit <http://www.alpha1awareness.org.uk>. A high alpha₁-antitrypsin level is associated with an acute phase response and does not require referral.

- **When should I test for Wilson's disease?**

If the patient is >40yr then Wilson's disease is very unlikely and measurement of caeruloplasmin is not indicated. If serum caeruloplasmin is < 0.2 g/L then measurement of 24hr urine copper excretion should be carried out in primary care (plain 24 hour urine). Serum copper measurements are not useful in diagnosing Wilson's disease. For more information, please visit <http://www.britishlivertrust.org.uk/liver-information/liver-conditions/wilsons-disease>

- **What causes a raised ferritin?**

Ferritin may be raised in a wide range of conditions including inflammation, obesity, alcohol excess, renal disease and liver disease. High ferritin with unexplained liver disease, infertility or fatigue may be suggestive of genetic haemochromatosis. If there is a family history of haemochromatosis, genetic screening is appropriate even if the TSat is normal.

See separate shared care hyperferritinaemia guideline for further guidance on the management of elevated ferritin.

For more information please visit <http://www.britishlivertrust.org.uk/liver-information/liver-conditions/haemochromatosis>

- **How do I interpret a raised immunoglobulin level?**

Serum immunoglobulins are not specific markers of liver disease. Please consider referral in patients with abnormal LFTs and a raised IgG, IgM or a polyclonal rise in immunoglobulins.

- Raised IgG is seen in Autoimmune Hepatitis
- Raised IgM associated with anti-mitochondrial Ab is common in Primary Biliary Cirrhosis
- A polyclonal rise in immunoglobulins may occur in patients with cirrhosis
- Raised IgA is common in steatotic liver disease (these patients could be assessed via the Community Liver pathway (see Appendix 2) unless other abnormalities in the NILS)

- **How do I interpret a positive smooth muscle antibody?**

Anti-smooth muscle antibody is non-specific and common in MASLD. If the IgG is normal and the ANA is negative it is not usually a concern. Autoimmune hepatitis should be considered, and the patient referred, if there is a concurrent raised IgG +/- positive ANA or no other explanation can account for the abnormal LFTs.

- **How do I interpret a positive anti-mitochondrial antibody?**

Positive anti-mitochondrial antibody (AMA) is strongly associated with primary biliary cholangitis and is often seen with an elevated IgM. If the patient with this has normal LFTs and is asymptomatic, it is recommended to monitor the LFTs annually and refer if/when these become abnormal and/or the patient develops symptoms or itch or fatigue.

- **How do I interpret an isolated raised alkaline phosphatase?**

A raised ALP with normal GGT is unlikely to be liver related. Check vitamin D and a bone profile including PTH, phosphate and calcium. The gut, pancreas, kidney and placenta are also alternative sources and a mild elevated ALP should be evaluated in the context of co-morbidities and underlying indication for LFT measurement.

- **How do I interpret an abnormal ultrasound?**

- *Indeterminate or suspicious liver mass* - urgent suspected cancer referral on HPB pathway

- *Liver Cyst*

- Simple and <10 cysts - no referral or monitoring required
- Simple and >10 cysts - risk of polycystic liver disease. Refer to hepatology.
- Simple large cyst (>8cm) and symptomatic with RUQ pain - refer to HPB surgery
- Complex cyst (septations, thick walled or biliary dilatation) - refer as urgent suspected cancer on HPB pathway

- *Haemangioma*

No referral or monitoring required if <5cm and typical features on ultrasound. Lesions >5cm or indeterminate lesions should be referred to hepatology for assessment.

- *Hepatomegaly*

Assess for risk factors for chronic liver disease and complete NILS. Manage results as per NILS box in pathway above.

- *Fatty liver or focal fat sparing*

Complete MASLD risk assessment via community pathway (see Appendix)

- *Gallbladder polyps*

See separate Shared Care Guidance on management

Background Information - Individual LFTs

Bilirubin

Bilirubin is mainly derived from the breakdown of haemoglobin during the normal turnover of red blood cells. It is transported in the blood stream bound to albumin and conjugated within the liver. Slight increases in bilirubin (up to 30 $\mu\text{mol/L}$) are not unusual and are not usually clinically significant.

Hyperbilirubinaemia may be either mainly due to conjugated or unconjugated bilirubin.

Causes of isolated unconjugated hyperbilirubinaemia:

- Gilbert's syndrome (bilirubin level usually < 70 $\mu\text{mol/L}$)
 - Common – affects up to 7% of the population
 - Haptoglobin not suppressed
 - Gilbert's Syndrome link <http://www.britishlivertrust.org.uk/liver-information/liver-conditions/gilberts-syndrome/>
- Haemolytic disease
 - Haptoglobin <200 mg/L is consistent with significant intra-vascular haemolysis
 - Reticulocytes and LDH may also be raised
 - Consider haematology referral
- Stress/fasting
- Drugs e.g. rifampicin, sulfonamides

Causes of isolated conjugated hyperbilirubinaemia:

- Drugs e.g. phenothiazines, sulfonamides and carbimazole
- Dubin-Johnson syndrome
- Rotor's syndrome

Alanine Transferase (ALT)

ALT is a cytosolic enzyme, which is expressed predominantly in liver cells and is used as a marker to assess liver cell damage. ALT < 120 IU/L are generally considered mild.

Common causes:

- Alcohol
- Viral hepatitis
- Steatosis
- Medications/toxins e.g. NSAIDs, antibiotics, statins, anti-epileptics, anti-TB drugs

Less Common causes:

- Autoimmune hepatitis
- Haemochromatosis
- Alpha₁-antitrypsin deficiency
- Wilson's disease

Non-hepatic causes of raised ALT (usually small rises, <120 U/L):

- Coeliac disease
- Strenuous exercise
- Muscle disease
- Endocrine disease e.g. hypo- and hyperthyroidism
- Regular use of protein supplements

Aspartate Aminotransferase (AST)

AST is expressed in the liver, as well as in the heart, skeletal muscle, kidneys, brain and red blood cells and therefore is not as liver specific as ALT. AST and ALT differ in their cellular location within the liver, as ALT is predominantly cytoplasmic and AST is present in both cytoplasm and mitochondria.

AST is not part of the initial LFT, but the ratio of AST to ALT may provide useful information about the possible cause of liver disease and the likelihood that the patient has advanced fibrosis.

Alkaline Phosphatase (ALP)

The two main sources of ALP are liver and bone, although there are also intestinal and placental isoforms. Elevations may be physiological or pathological.

Common causes for raised ALP:

Physiological

- Third trimester of pregnancy
- Adolescents, due to bone growth
- Benign, familial

Pathological

- Bile duct obstruction
- Primary biliary cirrhosis
- Primary sclerosing cholangitis
- Drug induced cholestasis, e.g. anabolic steroids
- Metastatic liver disease
- Bone disease e.g Pagets
- Heart failure

Gamma-glutamyl transferase (GGT)

GGT is a sensitive marker for hepatobiliary disease, but its use is limited by poor specificity.

Causes of raised GGT:

- Hepatobiliary disease (often with other liver enzyme abnormalities)
- Pancreatic disease
- Alcoholism
- Chronic obstructive pulmonary disease
- Renal failure
- Diabetes
- Myocardial infarction
- Drugs, e.g. carbamazepine, phenytoin and barbiturates and oral contraceptive pill

The use of GGT is in supporting a hepatobiliary source for other raised liver enzymes, e.g. ALP.

Raised ALP with normal GGT suggests ALP may be of bone rather than liver origin

Albumin

Albumin synthesis is an important function of the liver. When the functioning capacity of the liver decreases, falls in plasma albumin can be seen. However, there are many other causes of decreasing albumin levels.

Causes of low albumin:

- Decreased synthesis - severe liver disease, malabsorption, malnutrition, acute phase reaction
- Haemodilution - pregnancy, iv therapy, congestive cardiac failure, cirrhosis
- Altered distribution - injury, infection, inflammation, malignancy, cirrhosis
- Loss from body - skin (burns), gut (protein losing enteropathy) and renal (nephrotic syndrome)
- Increased catabolism - acute/chronic illness, malignancy, pregnancy

Resources

More information on liver diseases can be found at the British Liver Trust website:

<http://www.britishlivertrust.org.uk/liver-information/liver-conditions/>

Contacts

Duty Biochemist

01332 789383 (8am to 7pm, Monday-Friday)

On Call Consultant Biochemist

Via RDH switchboard, 01332 340131(24/7)

Reference

Newsome PN, Cramb R et al. Guidelines on the management of abnormal liver blood tests. Gut 2018; 67(1): 6-19.

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Laboratory (Derbyshire Pathology)	05.05.26	Ms S Knowles	Consultant Clinical Scientist
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Appendix: New Derbyshire CDC Fibroscan Pathway for Steatotic Liver Disease

The aim of the Fibroscan Pathway is to offer primary care direct access to diagnostic testing to promote earlier detection of liver disease, provide targeted brief intervention to patients, increase knowledge of liver disease in primary care and reduce routine referrals into secondary care services.

This pathway is designed for patients with suspected steatotic liver disease (due to metabolic factors or alcohol).

Inclusion Criteria

- Patient is 18 years old or over
 - FIB 4 1.3-3.25 if under 65 years; FIB 4 2.0-3.25 if 65 years or older
- OR**
- Harmful alcohol use: >35 units/week women, >50 units/week men

WITH

- Negative NILS (non-invasive liver screen) – normal results for:
 - TSH, Ferritin, Immunoglobulins, Coeliac Serology, Alpha-1-antitrypsin, Caeruloplasmin (<40yr only)
 - Hepatitis serology (Hep B surface Ag, Hep C Ab)
 - Negative liver autoantibodies (ANA, SMA, AMA, LKM), or titres <1:80
- Liver Ultrasound excluding significant structural liver disease (this does not include evidence of a fatty, coarse or bright liver)

Exclusion Criteria

- Fib 4 >3.25. These patients should be referred directly to secondary care hepatology as advanced chronic liver disease highly likely.
- Those that have an abnormal non-invasive liver screen (NILS). These patients should be referred to secondary care hepatology.
- Patients should not have any red flags (see Shared Care Guidelines for abnormal LFT in Adults) and should not require urgent onward referral to hepatology
- Pregnancy

The Fibroscan Pathway

For those referred to the CDC:

- The practitioner will carry out the Fibroscan.
- They will review and interpret the results according to set criteria.
- The practitioner will discuss the results and will include a targeted brief lifestyle intervention conversation with the patient.
- A recommendation will be made using standard criteria as to whether the patient requires onward referral into secondary care hepatology.
- In all cases, the detail of the appointment including the scan results and outcomes will be sent to Primary Care and the patient by letter so the patient record can be updated (via standard proforma).

How to refer

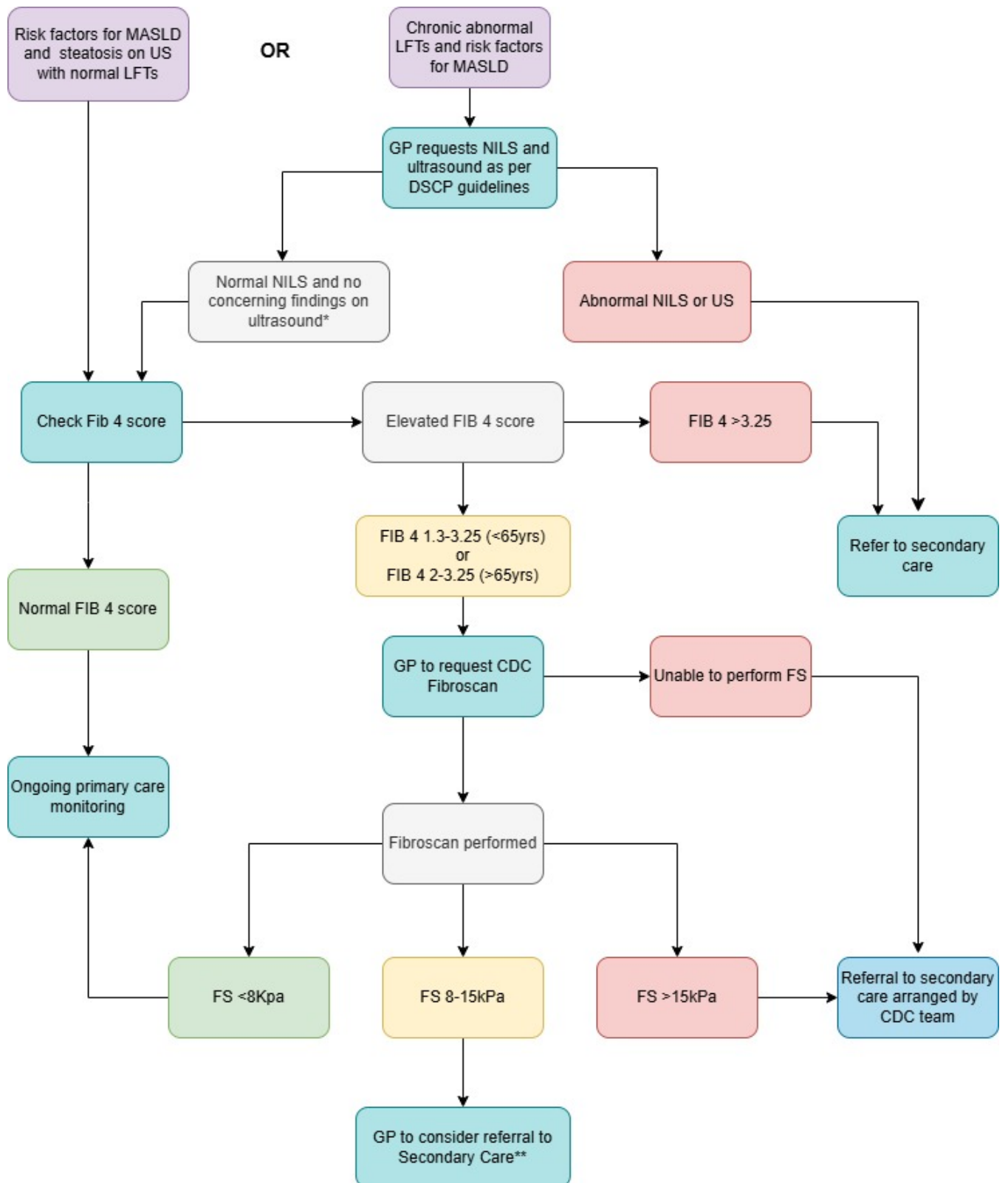
Primary care practitioner to complete referral for Fibroscan on ICE (South Derbyshire) or via email through Derbyshire Pathfinder (North Derbyshire).

What the patient can expect

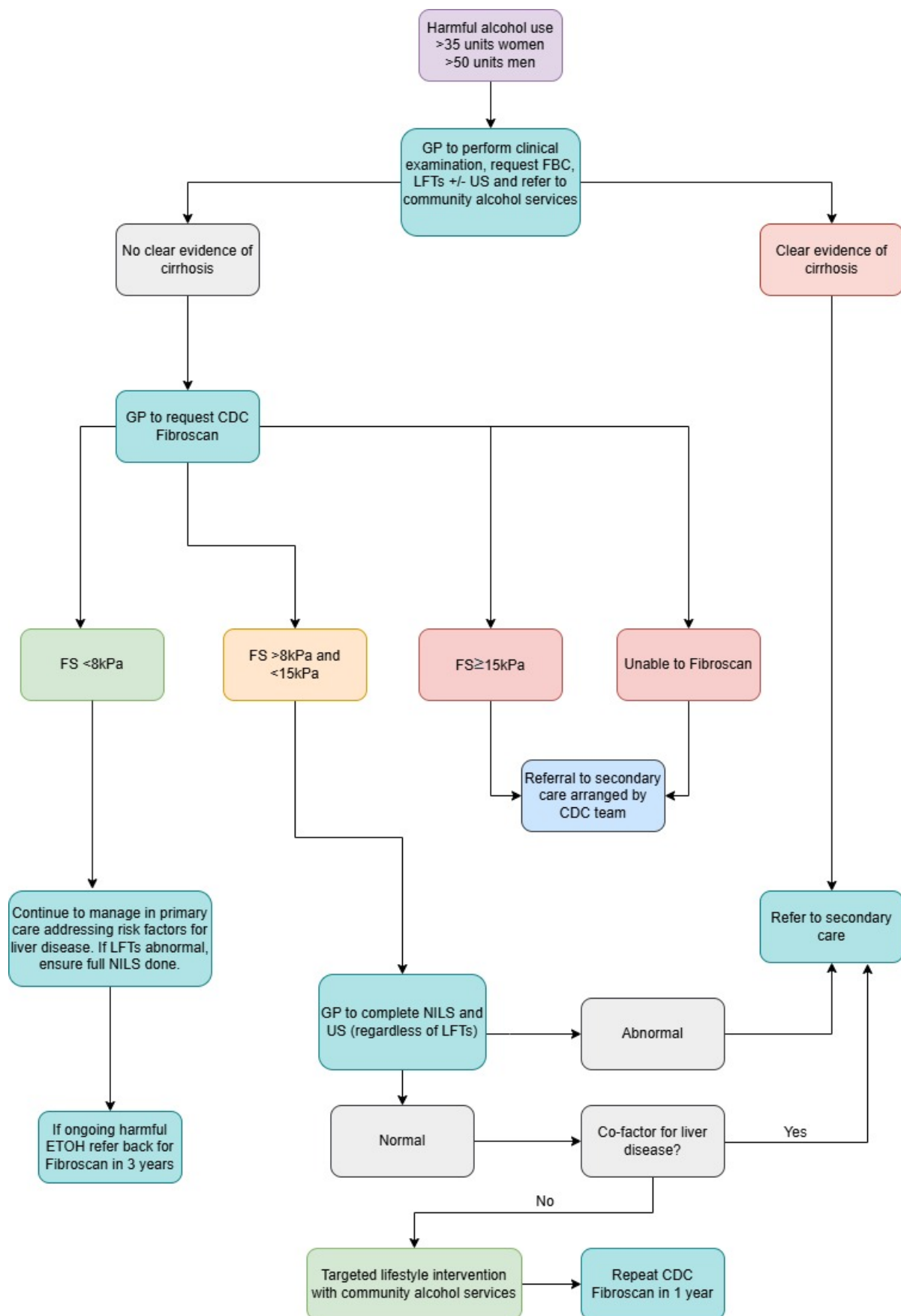
The patient will receive an appointment for a Fibroscan at the relevant nearest Community Diagnostic Centre within 30 days of referral. They will be advised to fast appropriately for the test. The results will be discussed with them on the day of the scan. The patient will receive a targeted brief lifestyle intervention conversation.

The patient's result will be communicated to Primary Care with advice for how to proceed.

Suspected MASLD pathway



Suspected alcohol-related liver disease pathway



Advice given to Primary Care with results

- Suspected MASLD
 - Fibroscan <8kPa : low risk, manage in primary care. Discharge with advice to GP:
 - Assess cardiovascular risk - QRISK2 and consider statin
 - Optimise diabetes and hypertensive control
 - Re-assess with Fib 4 measurement every 3 years if risk factors remain
 - Fibroscan 8-14.9kPa: Intermediate risk. GP to consider referral to Hepatology*
 - Fibroscan $\geq$ 15kPa: High-risk. Referral will be generated from the Fibroscan team to appropriate outpatient hepatology/gastroenterology services

- Harmful alcohol intake
 - Fibroscan <8kPa: manage in primary care. Discharge with advice to GP:
 - If ongoing consumption > 35/50 units per week - refer for repeat Fibroscan in 3 years and refer to community alcohol services if not already under their care
 - Fibroscan 8-14.9kPa: Intermediate risk. GP to complete investigations with NILS and ultrasound scan (if not already completed).
 - If NILS/ultrasound abnormal – refer to secondary care alongside referral to community alcohol services
 - If NILS/ultrasound normal
 - If co-factors for liver disease**, refer to secondary care
 - If no co-factors for liver disease, ensure referred to community alcohol service and request CDC Fibroscan in 1 year.
 - Fibroscan $\geq$ 15kPa: High-risk. Referral will be generated from the Fibroscan team to appropriate outpatient hepatology/ gastroenterology services

**Intermediate risk Fibroscan result in suspected MASLD*

Patient may have significant liver fibrosis or cirrhosis. Look for clinical features of cirrhosis (e.g. splenomegaly, thrombocytopenia). The decision to refer needs to factor in other co-morbidities or limited life expectancy, and consider whether the patient would be willing to accept further invasive investigation (such as liver biopsy) or is seeking further information about their liver disease and future risk.

Regardless, these patients will likely benefit from aggressive management of risk factors. Weight loss of 5-10% and reductions in alcohol to safe levels have been shown to arrest the progression from advanced liver fibrosis to cirrhosis. If not referred, consider an interval Fibroscan after 3 years.

*** Co-factors for liver disease*

These include obesity, Type 2 diabetes, dyslipidaemia, ischaemic heart disease, PCOS, and PIMZ phenotype for alpha-1-antitrypsin. The presence of one or more of these factors alongside alcohol raises the possibility of progression to advanced chronic liver disease. Such patients should be reviewed by Hepatology.

Unable to obtain Fibroscan (technically difficult/invalid result IQR>30%)

For patients in North Derbyshire: referral to Gastroenterology will be completed

For patients in South Derbyshire:

UHDB Hepatology will arrange an ELF test and communicate result to patient:

- If <9.8 , discharge back to primary care as low-risk fibrosis
- If ≥ 9.8 outpatient hepatology review will be arranged